

Bridging AI Solution Development Programme – Sabio Group

Health Fitness RAG project

Date: 15/06/2026 – 19/06/2026 (Monday – Friday)

Project Planning Document

Fitness & Health RAG Chatbot - Project Planning

1. Project Objective and Use Case

Objective

To develop an intelligent, interactive chatbot that provides personalized fitness and health guidance using Retrieval-Augmented Generation (RAG). The system aims to make fitness and health information accessible, actionable, and evidence-based for users of all fitness levels.

Use Case

The chatbot serves as a virtual fitness and health assistant that helps users with:

- × **Workout Planning:** Creating exercise routines for beginners to advanced levels
- × **Nutrition Guidance:** Providing meal planning, dietary recommendations, and nutritional information
- × **Sleep Optimization:** Offering evidence-based sleep hygiene practices and improvement strategies
- × **Stress Management:** Teaching stress reduction techniques and mindfulness practices
- × **Health Education:** Providing general wellness information and healthy lifestyle tips

Target Audience

- Individuals new to fitness looking for guidance
- People seeking to improve their health and wellness
- Those wanting evidence-based fitness information
- Users looking for motivational support and practical advice

Success Criteria

- Accurate retrieval of relevant information from the knowledge base
- Generation of helpful, encouraging, and evidence-based responses
- Positive user experience through an intuitive conversational interface
- Successful deployment on Hugging Face Spaces for public accessibility

2. Dataset and Data Sources

Knowledge Base Structure

The knowledge base consists of manually curated markdown files organized into six categories:

Category	Content	Files
Workouts	Exercise routines for beginner, intermediate, and advanced levels	beginner-workouts.md, intermediate-workouts.md
Nutrition	Meal planning, dietary guidelines, macro-nutrient information	meal-planning.md, dietary-guidelines.md
Sleep	Sleep hygiene, optimization techniques, sleep science	sleep-guide.md
Stress Management	Stress reduction techniques, mindfulness, relaxation	stress-management.md
Health	General wellness, daily habits, health metrics	general-health.md
Fitness Principles	Progressive overload, training fundamentals	progressive-overload.md

Data Characteristics

Format: Markdown (.md) with structured sections

Content Type: Evidence-based fitness and health information

Source: Curated from reputable fitness and health sources

Size: 8 documents, 42 chunks after processing

Language: English

Test Dataset

Format: JSONL (JSON Lines)

Purpose: Evaluate retrieval and answer quality

Content: 20+ test questions across all categories

Includes: Question, keywords, reference answer, category

Data Processing

Document Loading: All markdown files loaded using DirectoryLoader

Chunking: Documents split into 500-character chunks with 200-character overlap

Embedding: Text-embedding-3-large model from OpenAI

Vector Storage: Pinecone vector database with 3072-dimensional embeddings

Workflow Details

Query Flow:

1. User submits a question via Gradio UI
2. Query is resolved with conversation history context
3. Vector similarity search retrieves relevant chunks from Pinecone
4. Retrieved context is assembled and formatted
5. LLM generates response using context and conversation history
6. Response is returned to user with retrieved context for transparency

Data Ingestion Flow:

- Markdown files loaded from knowledge-base directory
- Documents split into overlapping chunks
- Chunks embedded using OpenAI embeddings
- Vectors upserted to Pinecone with metadata

4. Tools, Frameworks, and Technologies

Core Technologies

Component	Technology	Purpose
Language	Python 3.14+	Primary programming language
LLM	OpenAI GPT-4.1-nano	Response generation
Embeddings	OpenAI text-embedding-3-large	Document and query embedding
Vector DB	Pinecone	Vector similarity search
Framework	LangChain	RAG pipeline orchestration

Development Tools

Tool	Purpose
uv	Python package management and virtual environment
Git	Version control
GitHub	Repository hosting
Hugging Face Spaces	Application deployment
Pinecone Console	Vector database management

Infrastructure

Component	Specification
Vector Database	Pinecone Serverless (AWS us-east-1)
Embedding Model	text-embedding-3-large (3072 dims)
LLM	GPT-4.1-nano
Deployment	Hugging Face Spaces (Gradio SDK)
Storage	Pinecone (300MB+ capacity)

Using the Chatbot

Try asking questions like:

- "I'm new to fitness, where should I start?"
- "What should I eat before a workout?"
- "How can I sleep better?"
- "What are good stress management techniques?"
- "What is progressive overload?"

Evaluation

Running Tests

Metrics

Metric	Description	Target
MRR	Mean Reciprocal Rank	> 0.5
nDCG	Normalised Discounted Cumulative Gain	> 0.6
Keyword Coverage	Percentage of keywords found	> 80%
Accuracy	Factual correctness (1-5)	> 4.0
Completeness	Coverage of all aspects (1-5)	> 4.0
Relevance	Directness of answer (1-5)	> 4.0

Test Categories

- × Workout: Exercise routines and training
- × Nutrition: Diet and meal planning
- × Sleep: Sleep optimization
- × Stress: Stress management
- × Health: General wellness
- × Fitness Principles: Training fundamentals

Pinecone Index

Setting	Value
Index Name	fitness-health-main
Namespace	fitness-health
Dimension	3072
Metric	cosine
Cloud	AWS
Region	us-east-1

LLM Settings

Setting	Value
Model	gpt-4.1-nano
Temperature	0.3
Context Chunks	5
Chunk Size	500 characters
Chunk Overlap	200 characters

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- LangChain for the RAG framework
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- Gradio for the UI framework
- Hugging Face for deployment platform

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Disclaimer

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